Rewritten Chain of Thought for the Solution to the Two-Minus Water-Wave Amplitude

Rewritten Chain of Thought

This is a blind benchmark, so I am only allowed to read `prompt.md` and `OnShellBG.m` and must not touch any sibling file or any of the other solution folders. I want a closed-form A_n for $\sigma = (-1, -1, +1, \dots, +1)$, and the prompt hands me a real gift in its hint: the answer is a *piecewise homogeneous polynomial*, the kinematic space *decomposes into chambers*, and a *different homogeneous polynomial* holds on each chamber. I decide up front to take that literally — to hunt for the chamber walls and the per-chamber polynomials — rather than to fish for one global expression by brute fitting. I check the environment (`wolframscript`, a Mathematica kernel, and `sympy` are all available), then write a small driver that loads the BG *definitions* without running the file's own demo block, builds the two-minus sign vector, calls `MakeKinematics`, and feeds `BGAmplitude` in exact rational arithmetic.

The first runs settle the gross structure. The amplitude is purely imaginary: $A_5 = -2304\,i$, $A_6 = -\frac{295936}{11}i$, $A_7 = -\frac{4333568}{15}i$ on the file's own test points. A scaling test $\omega \to \lambda\omega$ shows $A_n \propto \lambda^{2(n-2)}$, i.e. homogeneous of degree $6, 8, 10$ at $n = 5, 6, 7$, so I write $A_n = -i\,P_n$ with P_n homogeneous of degree $2(n-2)$. And $n = 4$ refuses to compute: the on-shell conditions in this sector *force* $\omega_4 = -\omega_2$ and $\omega_1 = -\omega_3$, which lands an internal two-leg current on a propagator pole, so the recursion returns `Indeterminate` — a removable 0/0. I set $n = 4$ aside to recover later as a limit.

Because the external legs satisfy $|k_i| = \omega_i^2$ exactly, there is no sign ambiguity *per leg*; the only non-analytic content — the only thing that can make the answer piecewise — lives in the absolute values of *partial momentum sums* that mix the minus-legs with plus-legs inside the internal propagators. So I compute A_5 symbolically (it factors in 0.09 s), collect every `Abs` argument, and find that most are manifestly sign-definite (sums of squares) while the genuinely chamber-dependent walls are exactly the mixed subset sums like $\omega_2^2 - \omega_3^2$ and $\omega_2^2 - \omega_3^2 - \omega_4^2$. This is the prompt's chamber structure made concrete, and it tells me where to cut. Resolving the symbolic A_5 in specific chambers gives strikingly clean factored polynomials — in the chamber where ω_2 is the smallest, $P_5 = -16\,\omega_1\omega_2^5$; where it is the largest, $P_5 = -32\,\omega_1\omega_2\omega_3^2\omega_4^2$ — both carrying an overall $\omega_1\omega_2$, the product of the two minus-leg frequencies. So I write $P_n = -16\,\omega_1\omega_2\,R$ and chase R.

My first attempt to nail R goes wrong, and the way it goes wrong is instructive. I try to fit R as a polynomial in the free frequencies, grouping points by the sign-signature of all partial momentum sums. The fit comes back *inconsistent*, and for two reasons I only untangle in stages. First, P_5 is *rational*, not polynomial, in the free variables — it carries a $1/(\omega_2+\omega_3+\omega_4)$ from `MakeKinematics` solving ω_1, ω_5 — so I must divide it out. Second, when I broaden the sample to negative free frequencies I get 16 raw signatures, but only five fit cleanly as polynomials in the free *magnitudes*; the rest depend on the frequency *signs*, not just ω_i^2. My even-polynomial ansatz was simply too restrictive.

The breakthrough is to change the invariant. Instead of grouping by raw signature, I sort the legs by magnitude and label each by its sign, and I find that two points give the *same* R as a function of the sorted magnitudes $\mu_1 \le \dots \le \mu_n$ precisely when they share the same $\pm$ label sequence. The 16 signatures collapse, and only a handful of label sequences are even realizable, because momentum conservation $\sum_i \sigma_i\omega_i^2 = 0$ forbids most sign orderings. At $n = 5$ I find exactly four chambers, indexed by the sorted positions $\{p, q\}$ of the two minus-legs, with

$$\{1,5\}:\ R = \mu_1^2, \quad \{2,5\}:\ R = 2\mu_1\mu_2 - \mu_1^2, \quad \{3,4\}:\ R = 2\mu_1\mu_2,$$

$$\{3,5\}:\ R = 2\mu_3(\mu_1 + \mu_2) - (\mu_1^2 + \mu_2^2 + \mu_3^2).$$

I notice with some delight that the last one is $16\,(\text{Heron area})^2$ of a triangle with sides $|\omega_2|, |\omega_3|, |\omega_4|$, so triangle inequalities among magnitudes are going to be walls. The configuration $\{4,5\}$ never appears in four thousand points — it is unrealizable under *both* conservation laws. This is the hint's decomposition, found explicitly: one homogeneous polynomial per chamber, walls at subset-sum equalities.

$n = 6$ both confirms and complicates the picture. The same label-sequence grouping gives clean polynomials for some chambers ($\{1,6\} \to 2\mu_1^3$, $\{2,6\} \to 2\mu_1(\mu_1^2 - 3\mu_1\mu_2 + 3\mu_2^2)$, $\{4,5\} \to 12\mu_1\mu_2\mu_3$) but the label sequence is *not* a fine enough invariant: $\{3,6\}$ splits into sub-chambers across an extra wall, the triangle inequality $\mu_3 = \mu_1 + \mu_2$ among the three smallest legs. On one side $R = 6\mu_1\mu_2(2\mu_3 - \mu_1 - \mu_2)$; on the other $R = 2\,[\mu_3^3 - (\mu_3 - \mu_1)^3 - (\mu_3 - \mu_2)^3]$; and crucially R is *continuous* across the wall — both give $6\mu_1\mu_2(\mu_1 + \mu_2)$ at $\mu_3 = \mu_1 + \mu_2$. I write the difference between the two pieces as a single truncated-power correction,

$$R_{\{3,6\}} = 6\mu_1\mu_2(2\mu_3 - \mu_1 - \mu_2) - 2\,\left[\,\mu_1 + \mu_2 - \mu_3\,\right]_+^3,$$

and that is the moment the spline structure becomes unmistakable: a continuous piecewise polynomial with truncated-power corrections switching on at subset-sum walls. R depends only on $a \equiv$ min of the two minus-leg magnitudes and on the plus-leg magnitudes.

I do stumble around the right object before grabbing it. I generate $n = 6$ data in parallel (ten jobs, $\sim$5200 exact points), and at $n = 7$ where BG is $\sim$15 s per point I have far too little data to fit a degree-four polynomial per chamber. The per-chamber fits at first *look* mutually contradictory — the "6 versus 6.25" kind of mismatch — which I eventually trace to each chamber polynomial being defined only *modulo* momentum conservation, a red herring rather than a real disagreement. I try a naive overall constant $(n-4)!$, and at one point I talk myself into believing there is "no simple sum of truncated powers" at all. The honest fix is to stop fitting and run a decisive experiment: test the truncated-power ansatz

$$R \overset{?}{=} (n-4)! \sum_{S \subseteq \text{Plus}} (-1)^{|S|}\,\left[\,a - \sigma_S\,\right]_+^{n-3}, \qquad a = \min(\omega_1^2, \omega_2^2), \;\; \sigma_S = \sum_{i \in S} \omega_i^2,$$

directly against the raw amplitude at every collected point. It matches *all* of $n = 5$ (375/375) and *all* of $n = 6$ (4500/4500) exactly — but every $n = 7$ point is off by exactly $2/3$. That uniform factor is the tell: $(n-4)!$ gives $1, 2, 6$ at $n = 5, 6, 7$ whereas the data want $1, 2, 4$, i.e. the constant is 2^{n-5}, not $(n-4)!$ (they coincide at $n = 5, 6$ and first part ways at $n = 7$). Folding the constant into the prefactor gives the master formula

$$\boxed{A_n \;=\; i\,2^{n-1}\,\omega_1\omega_2 \sum_{S \subseteq \text{Plus}} (-1)^{|S|}\,\left[\,a - \sigma_S\,\right]_+^{n-3}}$$

with $a = \min(\omega_1^2, \omega_2^2)$, $\text{Plus} = \{\omega_3^2, \ldots, \omega_n^2\}$, and $\sigma_S = \sum_{i \in S} \omega_i^2$. The truncated powers carry the chamber decomposition automatically: a plus-subset S contributes iff $\sigma_S < a$, so the walls are exactly $\sigma_S = a$, and on each chamber the live set of S is fixed and A_n is a single homogeneous polynomial of degree $2(n-2)$ — a different polynomial on each chamber, precisely as the hint promised. The bare-monomial regime I started from, $R = a^{n-3}$, is just the chamber below every wall.

Once I read it as inclusion–exclusion the structure stops looking like a coincidence. With $k = n-2$ plus-legs the exponent $n - 3 = k - 1$ makes the sum a B-spline / Irwin–Hall density: $\sum_S (-1)^{|S|}\,[\,a - \sigma_S\,]_+^{k-1} = (k-1)!\,\left(\prod_{\text{Plus}} m_i\right) f_X(a)$ with $X = \sum_{\text{Plus}} m_i U_i$ and U_i uniform on $[0,1]$ — so R is, up to $\prod m_i$, the volume of a slice of the unit cube, the canonical positive-geometry "waterhedron" volume. That is why it is a spline and why it is continuous across every wall.

Two loose ends. For $n = 4$ the formula has exponent $n - 3 = 1$ and prefactor $2^{n-1} = 8$; the recursion's $0/0$ is recovered as a direction-independent limit, e.g. $\{s,t\} = \{2,3\}$ gives $-192\,i$, $\{1,5\}$ gives $-40\,i$, $\{7,3\}$ gives $-1512\,i$, each matching the formula exactly. And I verify the *single* boxed formula directly against `BGAmplitude` in exact rational arithmetic, where agreement is bit-exact and the relative error is identically zero, far below the 10^{-10} bar: 150 points each at $n = 5, 6$ spanning 4 and 12 distinct chambers, multi-chamber points at $n = 7$, the original three test points (reproduced exactly), an exhaustive earlier sweep of 5435 points with zero mismatches, an $n = 8$ spot-check ($A_8 = -\frac{33920}{21}i$), an independent Python reimplementation, and the $n = 4$ limits above. Sampling negative free frequencies on purpose pushes the kinematics into the other chambers, and the same formula holds there too — the truncated powers simply switch a different subset of terms on.

I want to be honest about the messy parts. A second solver was running on this same case and competing for Wolfram licenses; for a long stretch it starved my verification kernels, several of my background jobs hung mid-scan, and I spent real effort distinguishing my own stale processes from the foreign ones and re-launching clean runs rather than producing new mathematics. I also relabeled "chambers" twice — numeric activation fingerprints gave inflated counts (81 at $n = 5$) before I settled on the true combinatorial counts $4, 14, 59$ at $n = 5, 6, 7$ — and I corrected the overall constant once. None of that touched the final result: within the prompt's own kinematics, and in every chamber I could reach by sign-flipping the free frequencies, the boxed truncated-power formula reproduces `BGAmplitude` to exact equality at $n = 4, \ldots, 8$. The hint pointed straight at the piecewise structure, and the chamber-by-chamber discipline it suggested is what turned a pile of clean factored fragments into one closed form.